

Time series analysis is a branch of statistics and data science that focuses on studying data points collected or recorded over time in chronological order. Unlike other types of data, time series data has a temporal dependency, meaning that past values often influence future ones. This makes it especially useful for forecasting and understanding trends, seasonality, and cyclical patterns in real-world phenomena such as stock prices, weather conditions, sales demand, or website traffic. By applying techniques like decomposition, smoothing, and models such as ARIMA or exponential forecasting, analysts can extract meaningful insights and make predictions about future behavior based on historical patterns.